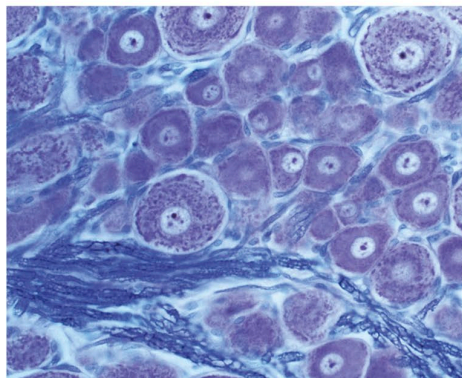
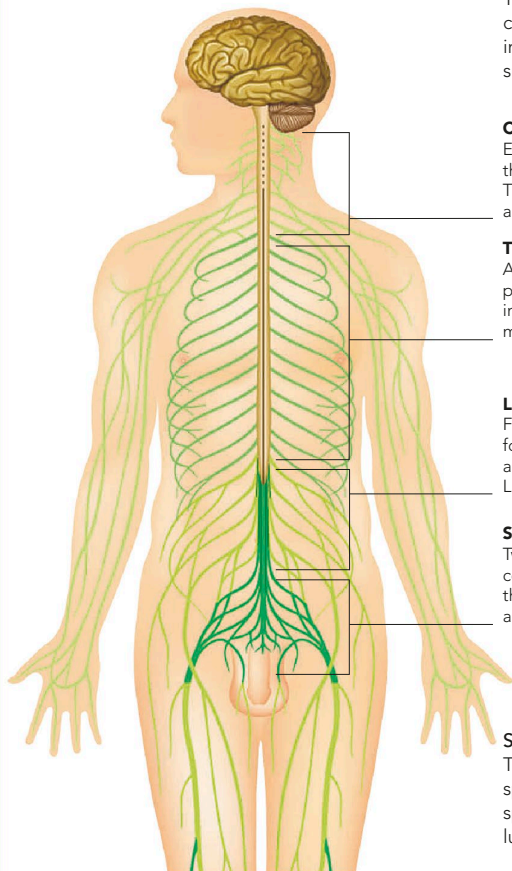


SPINAL NERVES

The 31 pairs of peripheral spinal nerves emerge from the spinal cord through spaces between the vertebrae (see p.99). Each nerve divides and subdivides into a number of branches; the dorsal branches serve the rear portion of the body, while the ventral branches serve the front and sides. The branches of one spinal nerve may join with other nerves to form meshes called plexuses, where nerves merge and intersect. Nerves leaving each plexus then go on to carry signals to and from that particular area of the body.



SPINAL NERVE GANGLION

This microscope image shows a section through a cluster of spinal nerve cells (ganglion), where nerve impulses are coordinated. Each neuron (purple) is surrounded by support cells (light blue).

Cervical region (C1–C8)

Eight pairs of cervical spinal nerves form two networks, the cervical (C1–C4) and brachial plexuses (C5–C8/T1). These run to the chest, head, neck, shoulders, arms, and hands, and to the diaphragm.

Thoracic region (T1–T12)

Apart from T1, which is considered part of the brachial plexus, thoracic spinal nerves are connected to the intercostal muscles between the ribs, the deep back muscles, and the abdominal muscles.

Lumbar region (L1–L5)

Four of the five pairs of lumbar spinal nerves (L1–L4) form the lumbar plexus, which supplies the lower abdominal wall and parts of the thighs and legs. L4 and L5 interconnect with the first four sacral nerves (S1–S4).

Sacral region (S1–S5)

Two nerve networks, the sacral plexus (L5–S3) and the coccygeal plexus (S4/S5/Co 1), send branches to the thighs, buttocks, muscles and skin of the legs and feet, and anal and genital areas.

SPINAL REGIONS

The organization and naming of the four main spinal nerve regions reflect the regions of the spine itself – cervical or neck, thoracic or chest, lumbar or lower back, and sacral or base of spine.